DATA SCIENCE LABORATORY MANUAL

Module V ? Advanced Data Science

Topic: Model Evaluation

OVERVIEW & LEARNING OBJECTIVES

Confusion matrix, precision, recall, F1-score, ROC-AUC curve, k-fold cross-validation, and bias-variance tradeoff.

KEY TOPIC CONCEPTS & METHODOLOGY

- Understanding Model Evaluation in modern Data Science workflows.
- Detailed breakdown: Confusion matrix, precision, recall, F1-score, ROC-AUC curve, k-fold cross-validation, and bias-variance
- Mathematical formulas, code examples, and laboratory execution guidelines.
- Common pitfalls, edge cases, and best practices in practical applications.

LABORATORY PRACTICE / IMPLEMENTATION NOTES

1. Follow standard Python and Data Science coding practices.
2. Document all exploratory findings and statistical hypotheses.
3. Keep datasets and outputs organized in your repository workspace.